

SOUJANYA CHAVAN

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EDUCATION

University of Southern California

Master of Science in Analytics (GPA: 3.54/4.00)

Los Angeles, CA

Aug. 2024 – May 2026

- **Coursework:** Predictive Analytics, Data Mining, Applied Generative Artificial Intelligence

A. P. Shah Institute of Technology, Mumbai University

Bachelor of Engineering in Computer Science (GPA: 3.67/4.00)

Mumbai, India

May 2020 – May 2024

- **Coursework:** Artificial Intelligence, Machine Learning, Database Management Systems, Deep Learning

TECHNICAL SKILLS

Generative AI & Agents: LangChain, LangGraph, multi-agent workflows, LLM-based workflows, tool calling, MCP orchestration, prompt engineering, RAG pipelines

LLM Evaluation & Retrieval: RAGAS, faithfulness, context precision, FAISS, ChromaDB, semantic retrieval, embeddings, RoBERTa, Hugging Face Transformers

Backend & APIs: Python, FastAPI, REST APIs, Pydantic, SQLAlchemy, PostgreSQL, MySQL, SQLite, RabbitMQ, Celery, Redis, Pandas, NumPy

Frontend & Languages: React, TypeScript, Next.js, Tailwind CSS, Python, Java, SQL

Cloud & MLOps: Docker, Git, GitHub Actions, CI/CD, AWS, Azure, pytest, Playwright, Claude Code, Codex

EXPERIENCE

Borealis Group Analytics

AI Engineer Intern

Blythewood, SC

July 2025 – December 2025

- Built a **LangChain multi-agent** Research Copilot end to end so analysts could ask plain-English questions and get evidence-backed answers, splitting planner, retrieval, SQL, Python-analysis, and synthesis agents into a debuggable application architecture.
- Wired **MCP orchestration** with typed tool schemas, role-restricted tools, and read-only SQL guardrails, cutting tool-routing failures from 22% to 9% across 300 queries.
- Raised answer **faithfulness** from 0.71 to 0.86 and context precision from 0.64 to 0.81 via a RAGAS benchmark of 250 analyst QA pairs.

USC Information Sciences

Machine Learning Research Assistant

Los Angeles, CA

October 2024 – May 2025

- Built the **React frontend** researchers used to explore 50K+ records a month with interactive filters for topic trends, ideology labels, confidence scores, and the highest-impact posts in an election influence study.
- Designed the **FastAPI backend** serving those views and cut dashboard load time from 2.8s to 1.1s with indexed PostgreSQL queries and cached aggregations.
- Fine-tuned **RoBERTa** with PEFT/LoRA lifting held-out accuracy from 78% to 86%, served over a RabbitMQ async pipeline scaling to 50K+ records/month.

Motleyscape

Machine Learning Engineer

India

August 2023 – June 2024

- Deployed Dockerized **ML services on Azure** that turned raw AR and VR session data and feedback into live signals on sentiment, usability issues, and feature adoption for product teams.
- Hardened the codebase into a maintained production repo with unit tests and **GitHub Actions CI**, cutting the broken deployments the team kept hitting.
- Lifted onboarding task success from 63% to 80% via **A/B tests** that settled which AR flow worked, then handed product a clear recommendation.

PROJECTS

Multi-Agent GenAI Business Analytics System

Oct 2025 – Dec 2025

- Built a **LangGraph** multi-agent app (Next.js + FastAPI) where one prompt turns a raw dataset into both a grounded data answer and a stakeholder deck, backed by a 4-provider LLM fallback chain for zero-downtime demos.
- Grounded the analyst agent with a persistent **ChromaDB** vector store and Pydantic-structured outputs, scored 0.86 retrieval precision on a 10-question Ragas benchmark.

Chat with SQL Database Using LangChain

Aug 2025 – Oct 2025

- Built a full-stack **text-to-SQL** app (React/TS + FastAPI) turning plain-English questions into safe SQL via Groq Llama 3.1, deployed as 3 containerized services across SQLite, PostgreSQL, and MySQL.
- Designed a **SQL guardrail** blocking 21 dangerous keywords plus injection, enforcing SELECT-only, read-only, single-statement execution.

RAG-Based Document Q&A System

Nov 2025 – Jan 2026

- Designed an end-to-end **RAG pipeline** with history-aware query reformulation, answering domain-specific questions from a large document set and curbing hallucinations by constraining answers to retrieved source text.

ACHIEVEMENTS

Honors in AI/ML (Academic Excellence): Awarded university-conferred Honors in Artificial Intelligence & Machine Learning, formally noted on the undergraduate degree.

Team Leadership: Led and coordinated the team through project delivery during the AI Engineer internship at Borealis Group Analytics.

Hiring Contribution: Interviewed an intern candidate for Borealis, contributing to the evaluation and final hiring decision.